

Automated Question Bank Generation System

From educational documents to an organized, searchable and reusable assessment engine.

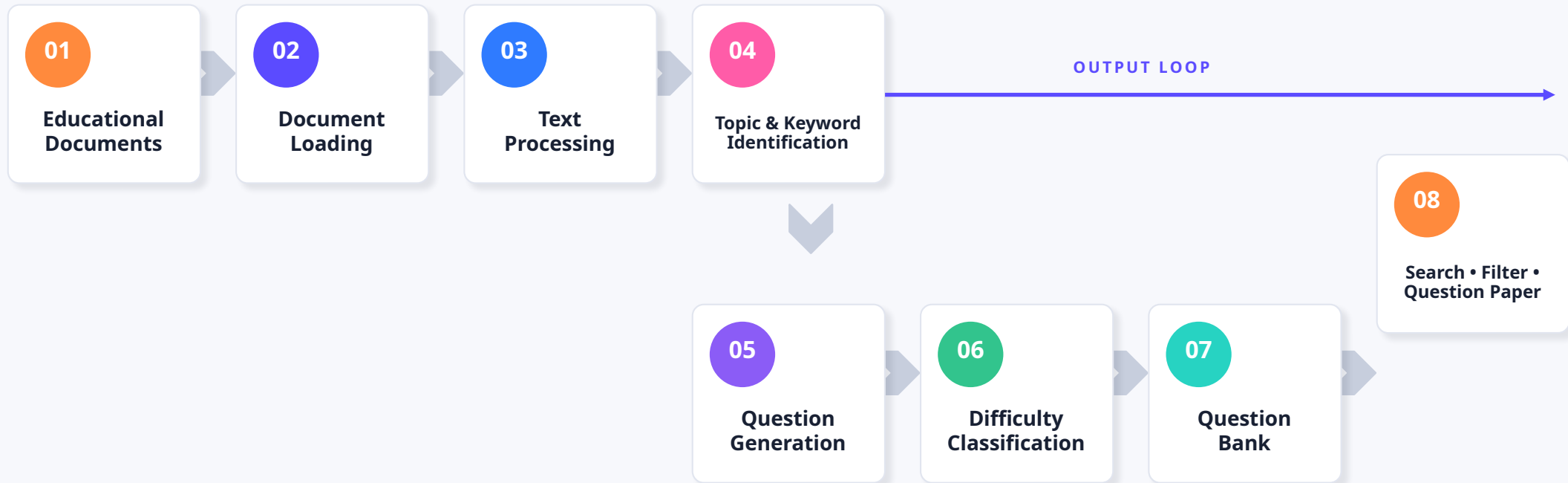
SECTION A5

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A clear path from content to assessment

Each stage transforms raw documents into a usable question bank.



STRUCTURED

REUSABLE

SEARCHABLE

The result feeds directly into retrieval and automatic paper creation.

From proposed concept to working prototype

A measurable shift from planning to implementation.

REVIEW 2 • PROPOSED

- Literature review
- Research gaps identified
- Automated system proposed
- Generation, organization and retrieval defined



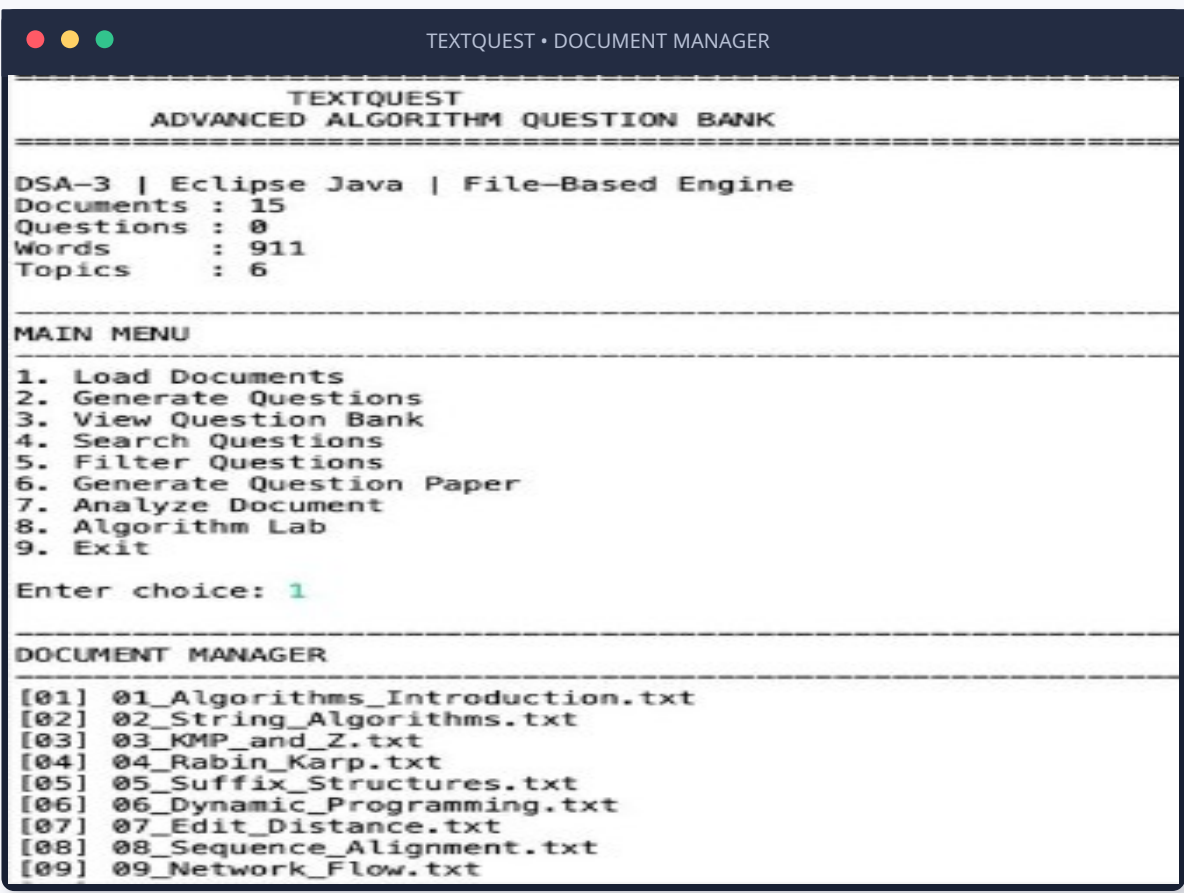
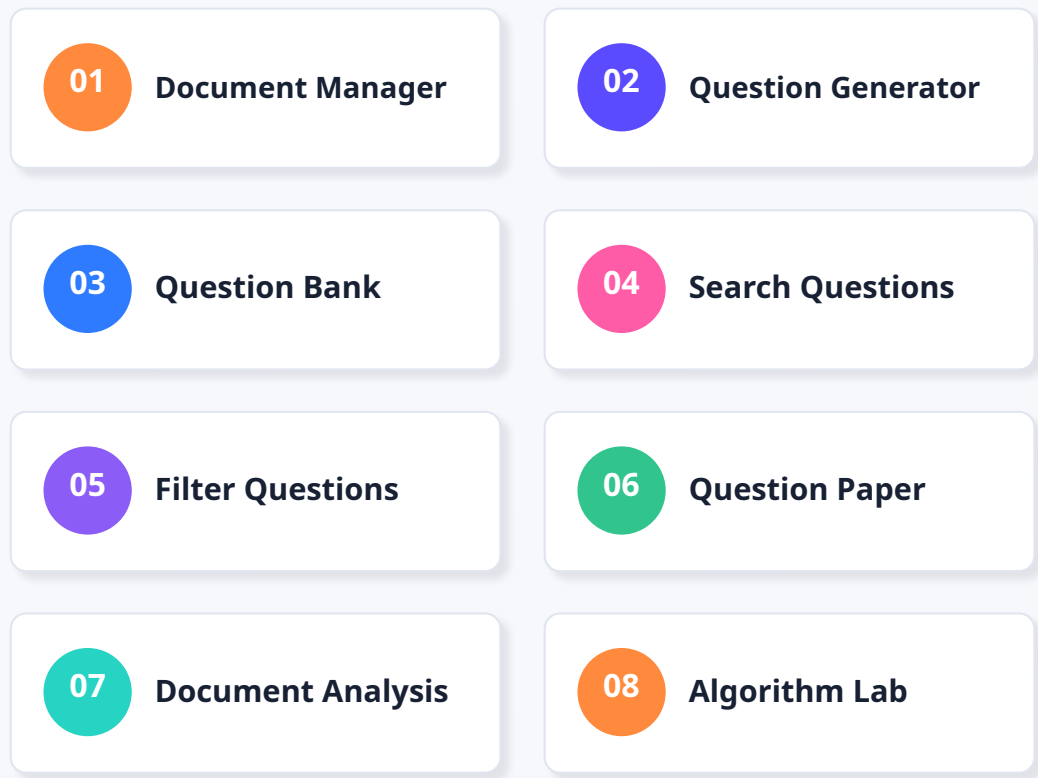
WORKING
PROTOTYPE

REVIEW 3 • IMPLEMENTED

- Java console prototype
- 15 documents • 911 words indexed
- 6 major topics identified
- Generation, search, filtering and paper creation
- Algorithm Lab implemented

Eight connected components

A modular design keeps loading, generation, retrieval and analysis distinct.



15 documents • 911 words • 6 topics

DSA algorithms powering the system

Core algorithm modules support search, analysis and experimentation.

STRING / TEXT

- 01 **KMP**
Pattern searching
- 02 **Z Algorithm**
Pattern analysis
- 03 **Rabin-Karp**
String matching
- 04 **Suffix Array**
Text indexing

DYNAMIC PROGRAMMING

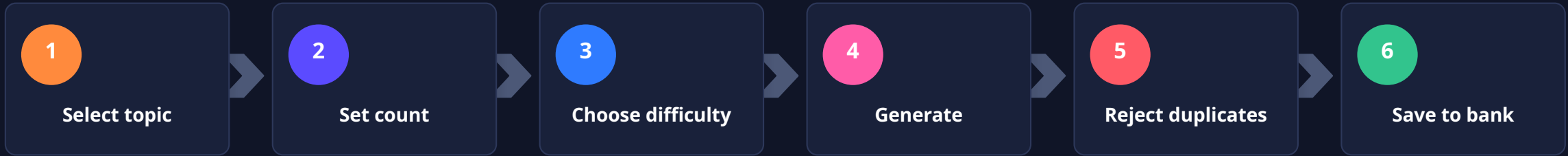
- 01 **Edit Distance**
Similarity & fuzzy matching
- 02 **Sequence Alignment**
Structured comparison

GRAPH / OTHER

- 01 **Max Flow**
Network analysis
- 02 **Vertex Cover**
Approximation
- 03 **Miller-Rabin**
Primality testing

Question generation & difficulty classification

A controlled six-step pipeline with duplicate protection.



DIFFICULTY LEVELS

EASY

MEDIUM

HARD

MIXED

```
QUESTION GENERATOR • TOPIC + COUNT + DIFFICULTY

QUESTION GENERATOR
1. All Documents
2. String Algorithms
3. Dynamic Programming
4. Network Flow
5. NP-Completeness
6. Randomized & Parallel Algorithms

Enter choice: 2
Question count: 5

Difficulty:
1. Easy
```

Question bank — search & filtering

Find the right question by keyword, topic, difficulty, algorithm or source.



BANK

Organized repository

Topic • Difficulty • Algorithm • Source



SEARCH

Keyword retrieval

Fast matching across generated questions



FILTER

Focused results

Refine by topic and difficulty

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SEARCH MENU

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MAIN MENU
-----
1. Load Documents
2. Generate Questions
3. View Question Bank
4. Search Questions
5. Filter Questions
6. Generate Question Paper
7. Analyze Document
8. Algorithm Lab
9. Exit

Enter choice: 4
Keyword: algorithm
  
```

```

MATCHING QUESTIONS • KEYWORD: ALGORITHM

Enter choice: 4
Keyword: algorithm
[Q001] Easy | String Algorithms
For S = "aabxaabxcaabxaabxay", construct the complete Z-array and identify all posit
Algorithm: Z Algorithm
Source : 01_Algorithms_Introduction.txt

[Q002] Easy | String Algorithms
Construct the suffix array of "banana" and list the suffixes in sorted order. Report
Algorithm: Suffix Array
Source : 02_String_Algorithms.txt

[Q003] Easy | String Algorithms
Compare KMP and Rabin-Karp for text "AAAAABAAAAAB" with pattern "AAAB". State which
Algorithm: String Comparison
Source : 03_KMP_and_Z.txt

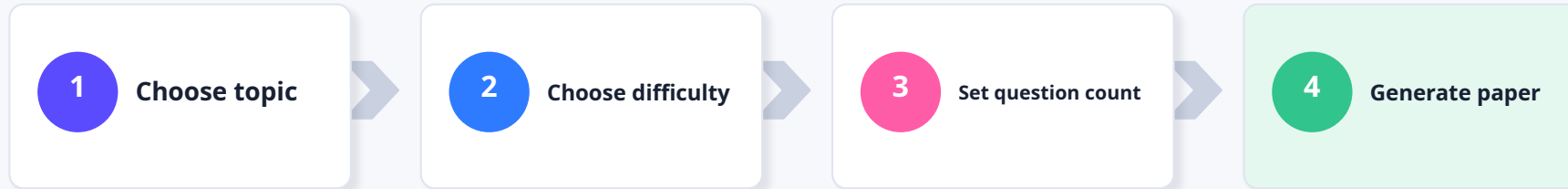
[Q004] Easy | String Algorithms
Construct the Z-array for "aaaaa" and explain why the values form a descending seque
Algorithm: Z Algorithm
Source : 04_Rabin_Karp.txt

[Q005] Easy | String Algorithms
Build the suffix array for "abracadabra" and state how suffix ordering supports fast
Algorithm: Suffix Array
Source : 05_Suffix_Structures.txt

Matches: 5
Press Enter to continue...
  
```

Automatic question paper generation

A simple selection flow converts the question bank into an assessment-ready paper.



QUESTION SELECTION

- Specify the required number
- Select topic and difficulty
- Draw from the saved question bank
- Produce a ready-to-use paper

```

FILTERED QUESTION SELECTION

Enter choice: 5
Topics:
1. String Algorithms
2. Dynamic Programming
3. Network Flow
4. NP-Completeness & Approximation
5. Randomized Algorithms
6. Parallel Algorithms
Topic: 1
Difficulty:
1. All
2. Easy
3. Medium
4. Hard
Difficulty: 1
[Q001] Easy | String Algorithms
For S = "aabxaabxcaabxaabxay", construct the complete Z-array a
Algorithm: Z Algorithm
Source : 01_Algorithms_Introduction.txt
```

Implementation evidence

Prototype results demonstrate end-to-end execution.

15

Documents

911

Words indexed

6

Major topics

5

Demo questions

```

SUCCESSFUL DOCUMENT LOADING

Enter choice: 0
Loading documents...
[OK] 15 documents loaded
[OK] 911 words indexed
[OK] 6 major topics identified

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MAIN MENU
-----
1. Load Documents
2. Generate Questions
3. View Question Bank
4. Search Questions
5. Filter Questions
6. Generate Question Paper
7. Analyze Document
8. Algorithm Lab
9. Exit
Enter choice: 2
    
```

```

GENERATED QUESTION OUTPUT

Enter choice: 5
Topics:
1. String Algorithms
2. Dynamic Programming
3. Network Flow
4. NP-Completeness & Approximation
5. Randomized Algorithms
6. Parallel Algorithms

Topic: 1
Difficulty:
1. All
2. Easy
3. Medium
4. Hard

Difficulty: 1

[Q001] Easy | String Algorithms
For S = "aabxaabxcaabxaabxay", construct the complete Z-array a
Algorithm: Z Algorithm
Source : 01_Algorithms_Introduction.txt
    
```

WORKING PROTOTYPE

NEXT CHAPTER

From a console prototype to an intelligent assessment platform.

TEXTQUEST

LIMITATIONS, FUTURE SCOPE & CONCLUSION

CURRENT LIMITATIONS

- Console-based interface
- Depends on supplied corpus
- Question quality can improve
- Difficulty estimation can improve

FUTURE SCOPE

- Web-based interface
- Larger, diverse datasets
- Stronger NLP generation and classification
- Multi-format question paper export

“From educational content to a structured, searchable question bank.”